

Y2 Electromagnetism 2

Non-assessed problem answers 1 (weeks 1–2)

Revision – Ideal conductors – Electric dipoles

1) $Q = -5.4 \times 10^5 \text{ C}$.

2) The electric field and its asymptotic behaviour for $|z| \gg R$ are

$$E_z(z) = \frac{Q}{4\pi\epsilon_0} \cdot \frac{z}{(z^2 + R^2)^{3/2}} \xrightarrow{|z| \gg R} \frac{Qz}{4\pi\epsilon_0|z|^3}.$$

Asymptotically, we obtain the Coulomb's law. The field is strongest at $z_0 = \pm R/\sqrt{2}$, with a magnitude of $E(z_0) = Q/(6\sqrt{3}\pi\epsilon_0 R^2)$.

3) The electric field is $E_z(z) = \frac{\sigma}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + R^2}} \right)$.

Far from the disc ($|z| \gg R$), $E_z(z) = Q/4\pi\epsilon_0 z^2$, i.e. the field of a point charge.

Near the disc ($|z| \ll R$), $E_z(z) = \sigma/2\epsilon_0$, i.e. the field of an infinitely large plane.

4) $\rho(r) = 2\epsilon_0 E_0/r$; $Q = 4\pi\epsilon_0 E_0 R^2$.

5) The magnetic field and its asymptotic behaviour for $|z| \gg R$ are

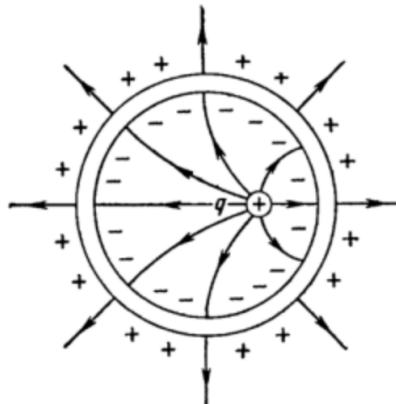
$$B_z(z) = \frac{\mu_0 I R^2}{2(z^2 + R^2)^{3/2}} = \frac{\mu_0 m}{2\pi(z^2 + R^2)^{3/2}} \xrightarrow{|z| \gg R} \frac{\mu_0 m}{2\pi|z|^3},$$

where $m = \pi R^2 I$ is the magnetic moment of the loop. The field is strongest in the plane of the loop ($z_0 = 0$), with a magnitude of $B(z_0) = \mu_0 I/2R$. Asymptotically for $|z| \gg R$, we obtain the magnetic dipole field as expected.

7) The force is attractive, $\delta F/\delta l = \mu_0 I^2/2\pi r = 0.002 \text{ N/m}$.

8) Use the definitions of the ∇ , gradient, divergence and curl operators.

9) For the second (less trivial) case,



10) It can be shown by a direct calculation that the electrostatic potential due to the charges Q and q at any point of the sphere is $V = 0$, which is sufficient to prove that q is the electrostatic image of Q . There is also an elegant geometrical proof involving similar triangles. The surface charge densities are

$$\sigma_A = \frac{Q}{4\pi} \cdot \frac{1 - d/R}{(d + R)^2} < 0; \quad \sigma_B = -\frac{Q}{4\pi} \cdot \frac{1 + d/R}{(d - R)^2} < 0.$$

11)
$$E_A = \frac{3p}{4\pi\epsilon_0 R^3}, \quad E_B = 0.$$

12) The electric field is

$$\vec{E}_A(r) = \frac{\vec{p}}{2\pi\epsilon_0 r^3}, \quad \vec{E}_B(r) = -\frac{\vec{p}}{4\pi\epsilon_0 r^3}.$$

13) In the first case, on the horizontal (vertical) axis, the field is vertical (horizontal), with a magnitude of

$$E(r) = \frac{3qa^2}{4\pi\epsilon_0 r^4}.$$

In the second case, the field is the same as above on the vertical axis, and twice as strong on the horizontal axis.

14) See lecture 4.
